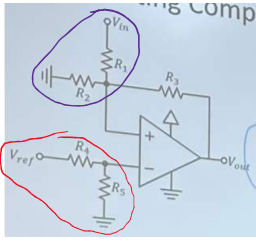
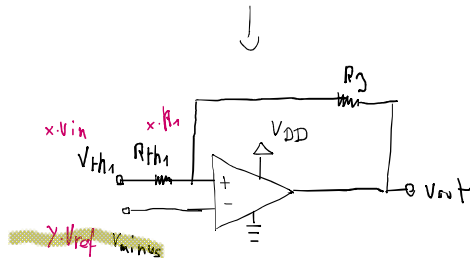


Vbatt H	4.2	+50mV	[3.8V]
Vbatt L	2.8V	+100mV	[3.0V]



V_{th1} R_{th1}
 V_{minus}



$$x = \frac{R_2}{R_1 + R_2}$$

$$y = \frac{R_5}{R_4 + R_5}$$

$$z = \frac{R_7}{R_1 + x \cdot R_1}$$

$$V_+ = \frac{R_2}{R_3 + x \cdot R_1} \cdot x \cdot V_{in} + \frac{x \cdot R_1}{R_3 + x \cdot R_1} \cdot V_{out}$$

$$V_+ = z \cdot x \cdot V_{in} + (1 - z) V_{out}$$

$$V_{out} = 0 \text{ V}$$

$$V_{in} = \nearrow$$

$$V_+ = V_-$$

$$y V_{ref} = x z V_{inH}$$

$$V_{out} = V_{DD}$$

$$V_{in} = \searrow$$

$$V_+ = V_-$$

$$y V_{ref} = x z V_{inL} + (1 - z) V_{DD}$$

$$\Rightarrow z = \frac{y}{x} \cdot \frac{V_{ref}}{V_{inH}} \rightarrow y V_{ref} = y \frac{V_{ref}}{V_{inH}} V_{inL} + \left(1 - \frac{y}{x} \frac{V_{ref}}{V_{inH}}\right) V_{DD}$$

$$x = 1$$

$$y V_{ref} = y \frac{V_{ref}}{V_{inH}} \cdot V_{inL} + \left(1 - y \frac{V_{ref}}{V_{inH}}\right) V_{DD}$$

$$y (V_{ref} \cdot V_{inH} - V_{ref} \cdot V_{inL} + V_{ref} \cdot V_{DD}) = V_{inH} \cdot V_{DD}$$

$$y = 1$$

$$V_{ref} = \frac{V_{ref} \cdot V_{inL}}{V_{inH}} + \left(1 - \frac{1}{x} \frac{V_{ref}}{V_{inH}}\right) V_{DD}$$

$$x \cdot V_{inH} \cdot V_{ref} = x V_{ref} V_{inL} + x V_{DD} \cdot V_{inH} - V_{ref} V_{DD}$$

$$x \cdot (V_{ref} V_{inL} + V_{DD} V_{inH} - V_{inH} V_{ref}) = V_{ref} V_{DD}$$

$$R_3 = 1 \text{ e } 6$$

$$R_1 = R_3 \frac{1 - z}{x z}$$

$$R_2 = R_1 \frac{x}{1 - x}$$

$$R_4 = 1 \text{ e } 6$$

$$R_5 = R_4 \frac{y}{1 - y}$$

$$V_{inH} = \frac{y V_{ref}}{x z}$$

$$V_{inL} = \frac{y V_{ref} - (1 - z) V_{DD}}{x z}$$